

# An Interesting Rearrangement

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This exploration features another rearrangement with an interesting transition-state structure.<sup>1</sup> We will discuss the observation of general base and specific acid catalysis in the mechanism for the monocyclic rearrangement of hydrazones of oxadiazoles. This reaction has the synthetic utility of allowing for a conversion between two types of heterocyclic rings. The authors report data for a series of phenyl-substituted hydrazones, allowing for a Hammett analysis of the system. Conclusions can be drawn about the proton transfer in the transition states for the acid catalyzed, base catalyzed and uncatalyzed pathways.

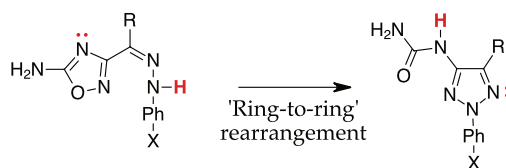
Heterocyclic compounds are important in medicinal chemistry.<sup>2,3</sup> Developing methods for their synthesis and manipulation is a goal of many research groups. The reaction being investigated here is an example of a monocyclic rearrangement of a heterocycle and the general form of this reaction is shown in figure 1. The reaction works best with a 1,2-oxazole ring in the starting material. Note the position of the O and N atoms in the general scheme.

## Part 1: The pH-Rate Profile

In this section we will plot the pH-Rate profiles for the series of reactants. A model that predicts the value of  $k_{obs}$  vs pH will then be constructed.

### The Reaction

The reaction investigated in this paper<sup>1</sup> would allow for the synthesis of a substituted 1,2,3-triazole from a hydrazone of a 1,2,4-oxadiazole as shown in figure 2. This change requires the removal of one proton and the arrival of another and these proton transfer steps give the opportunity for acid/base catalysis.



### Rate vs. pH

The authors present a pH-rate profile that has three distinct regions. In acidic conditions, the authors observe an acid-catalyzed pathway where they propose protonation at the 4-nitrogen of the oxadiazole ring. The protonated oxadiazole is more electrophilic and attack of the hydrazone nitrogen is faster

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<sup>1</sup> "On the Rearrangement in Dioxane/Water of (Z)-Arylhydrazones of 5-Amino-3-benzoyl-1,2,4-oxadiazole into (2-Aryl-5-phenyl-2H-1,2,3-triazol-4-yl)ureas: Substituent Effects on the Different Reaction Pathways." F. D'Anna, V. Frenna, et al., *J. Org. Chem.*, 2006, 71, 5616-5624. <https://doi.org/10.1021/jo0605849>

<sup>2</sup> "Oxadiazoles in Medicinal Chemistry." J. Boström, A. Hogner et al., *J. Med. Chem.*, 2012, 55, 1817-1830. <https://doi.org/10.1021/jm2013248>

<sup>3</sup> "Triazoles and Their Derivatives: Chemistry, Synthesis, and Therapeutic Applications." M.M. Matin, P. Matin et al., *Frontiers in Molecular Biosciences*, 2022, 9, 864286, <https://10.3389/fmolb.2022.864286>



Figure 1: The "ring to ring" rearrangement scheme

Figure 2: The rearrangement reaction. The substituted phenyl ring provides the Hammett series. Protons and lone pairs involved in acid/base proton exchange are highlighted in red.

than in the absence of protonation. In basic conditions, the authors observe general base catalysis where deprotonation of the hydrazone nitrogen is required to assist its nucleophilic attack on the oxadiazole ring. During this rate-determining step, partial removal of the proton by a general base (be it hydroxide or buffer) increases the nucleophilicity and the speed the rearrangement reaction. In more neutral conditions, the reaction is also observed to occur without catalysis where the ring closes in a mechanism involving only the water in the solvent. A pH-rate profile of one of the reactants, **1f** (where  $R = \text{Cl}$ ), is shown in figure 3 and displays the three distinct dependencies on acidity.

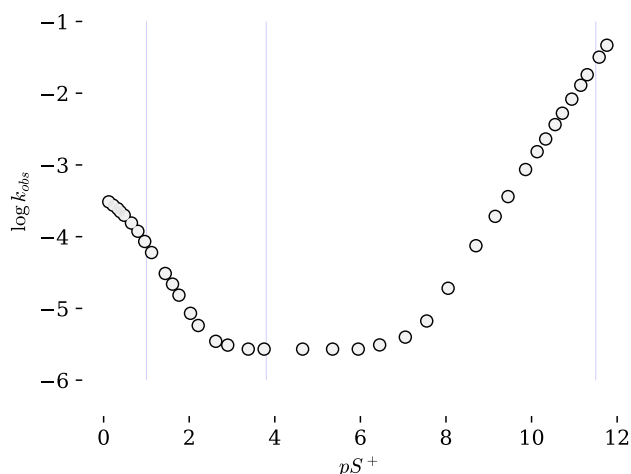
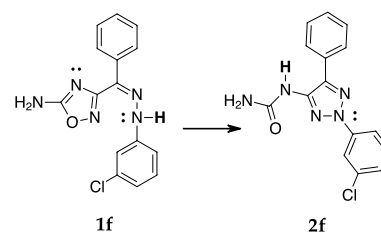


Figure 3: pH-rate profile for rearrangement of **1f** in 1:1 dioxane/water. Vertical lines highlight the values of  $pS^+ = 1.0, 3.8$  and  $11.5$ .

Code for this plot can be explored via [Google Colab](#)



The authors chose to investigate substituent effects on the rate of reaction at  $pS^+ = 1.0, 3.8$  and  $11.5$  where the rates of each mechanism are most easily revealed. Let us consider the three mechanisms observed in the pH-rate profile. We will start by stating rate laws for each and then creating a combined rate law that will predict the rate at any given pH value.

### Hydroxide Catalysis

In general-base catalysis, the proton is transferred during the rate-determining step. Any base can participate, but stronger bases will have larger rate constants. The hydroxide ion or the basic form of any buffers may act as the base. Increasing buffer concentration at a given value of  $pS^+$  will give a faster rate as there will be more contribution to the total rate from the buffer. One can measure the rate at several buffer concentrations and then extrapolate back to zero buffer concentration to isolate the rate for the hydroxide catalysis.

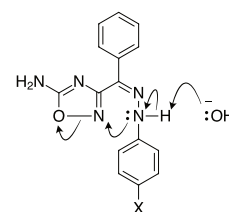
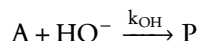


Figure 4: General-base catalysis by hydroxide.

The reaction scheme for this mechanism is a simple second-order reaction.



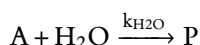
The rate law is stated in eq. 1.

$$\frac{d}{dt}[P] = k_{\text{OH}}[\text{OH}^-][A] \quad (1)$$

### The Water Rate

As we move to more acidic conditions, we reach a point where the concentration of hydroxide is very small and the rate due to general-base catalysis by the solvent<sup>4</sup> becomes the fastest observed rate. The rate will be a constant because the water solvent is acting as the base.

The reaction scheme for this mechanism is a simple second-order reaction.



The rate law is a first-order rate law, because the concentration of water is constant. This rate law is stated in eq. 2.

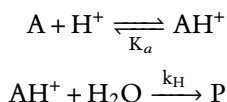
$$\frac{d}{dt}[P] = k_{\text{H}_2\text{O}}[A] \quad (2)$$

This rate is often referred to as the "uncatalyzed" rate because it is the background rate before any acid or base is added. However, in this case, it is general base catalysis by solvent.

### Acid Catalysis

In stronger acid conditions, we observe the rate dominated by a reaction that is catalyzed by acid. The reactant is being protonated, and this protonated intermediate reacts much more quickly with water (and water is the only base present in acidic conditions). The intermediate is formed before the rate-determining step as an acid equilibrium and the amount of protonated reactant depends on the protonating power of the solution, pH (or  $\text{pS}^+$  in this case). This is an example of specific catalysis by acid.

The reaction scheme for this mechanism is a simple second-order reaction with an acid equilibrium preceding the rate-determining step.



The rate law is a first-order rate law and is stated in eq. 4. Here the solvent still acts as a general base with the much more reactive protonated reactant.

<sup>4</sup> Water is a very weak base, but it is still a base, and it is present at a very high concentration as part of the solvent mixture.

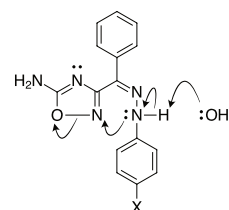


Figure 5: General-base catalysis by water.

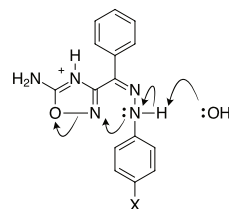


Figure 6: Specific acid catalysis.

$$[\text{AH}^+] = \frac{[\text{H}^+]}{[\text{H}^+] + K_a} [\text{A}_t] \quad (3)$$

$$\frac{d}{dt} [\text{P}] = k_H [\text{AH}^+] \quad (4)$$

### The Total Rate

With the power of modern computers we can fit a model for the pH-rate profile and obtain values for all four parameters,  $pK_a$ ,  $k_{OH}$ ,  $k_{H_2O}$  and  $k_H$ . Is there enough information to accurately determine  $pK_a$ ? We will soon see.

The rate law that defines the model is the sum of the rates of the three reactions.

$$\frac{d}{dt} [\text{P}] = k_{OH} [\text{OH}^-] [\text{A}] + k_{H_2O} [\text{A}] + k_H [\text{AH}^+] \quad (5)$$

We could easily fit the experimental data in each section of the graph where a given rate dominates, and determine the rate constants in a piecewise fashion. In strongly acidic conditions, we can see a slight curvature of the rate profile at as we approach  $pS^+ < 0$  in figure 3 on page 2. The  $pK_a$  value of the reactant seems to be close to a value of zero. We will be able to obtain values for  $pK_a$  in most cases, although the precision will likely be low.

We can state that  $[\text{A}] = [\text{A}]_t$  in the regions where the  $k_{OH^-}$  and  $k_{H_2O}$  terms are dominant. We can calculate  $[\text{AH}^+]$  in terms of  $K_a$  and  $[\text{H}^+]$  as  $[\text{AH}^+] = ([\text{H}^+]/K_a + [\text{H}^+]) [\text{A}_t]$ . Finally, we know that  $[\text{OH}^-] = K_w/[\text{H}^+]$ . We can substitute these into eq. 5 to obtain eq. 6.

$$\frac{d}{dt} [\text{P}] = k_{OH} \frac{K_w}{[\text{H}^+]} [\text{A}]_t + k_{H_2O} [\text{A}]_t + k_H \frac{[\text{H}^+]}{K_a + [\text{H}^+]} [\text{A}_t] \quad (6)$$

$$\frac{d}{dt} [\text{P}] = \left( k_{OH} \frac{K_w}{[\text{H}^+]} + k_{H_2O} + k_H \frac{[\text{H}^+]}{K_a + [\text{H}^+]} \right) [\text{A}_t]$$

and so...

$$\frac{d}{dt} [\text{P}] = k_{obs} [\text{A}]_t$$

where...

$$k_{obs} = k_{OH} \frac{K_w}{[\text{H}^+]} + k_{H_2O} + k_H \frac{[\text{H}^+]}{K_a + [\text{H}^+]} \quad (7)$$

The  $k_{A,R}$  values reported in the paper are  $k_{obs}$  (the authors used the term  $k_{A,R}$ )<sup>5</sup> and the data sets are therefore  $k_{obs}$  vs.  $pS^+$ . The value of  $pS^+$  is the effective pH value. So  $10^{-pS^+}$  would be the activity of  $\text{H}^+$ . The solution is 50% dioxane and thus the activity and concentration may not be exactly equivalent, but it will do for our analysis.

<sup>5</sup> By inspecting eq. 7, we can assign the components of the equation for  $k_{obs}$  to the pH region where they are the dominant contributor to the overall rate. We can then interpret the observed rate constants presented by the authors using our model.

$$k_{A,R_{pH\ 11.5}} = k_{OH} [\text{OH}^-] = k_{OH} \frac{K_w}{[\text{H}^+]}$$

$$k_{A,R_{pH\ 3.8}} = k_{H_2O}$$

$$k_{A,R_{pH\ 1.0}} = \frac{k_H}{K_a} [\text{H}^+]$$

### Summary

We have examined a pH-rate profile for the rearrangement and hypothesized a scheme that involves three different reactions. We proposed rate laws for each reaction and totalled them up to make a model that predicts rate with given values of pH ( $\text{pS}^+$  in this case.)

### A Note on the Rate Law Model

If we wanted to be precise, We could express the rate law without using the assumption that  $[A] = [A]_t$  in the neutral and basic regions of the plot. The observed rate constant would then be expressed as...

$$k_{obs} = \left( k_{OH} \frac{K_W}{[H^+]} + k_{H_2O} \right) \frac{K_a}{K_a + [H^+]} + k_H \frac{[H^+]}{K_a + [H^+]} \quad (8)$$

Compare Eq. 8 with Eq. 7 [on the previous page](#). Convince yourself that when  $[H^+]$  is large, only the right-hand term in Eq. 8 is significant, and that when  $[H^+]$  is large small, the  $K_a/K_{a+[H^+]}$  term vanishes. Or you can alter the code in the curve fits to use the full equation and see that it makes no difference.

## Part 2: Reproducing the Hammett Plots

We have been exploring a rearrangement reaction that features general-base and specific-acid catalysis.<sup>1</sup> In this section I will use the authors' reported values for the rates observed at pH = 11.5, 3.8, and 1.0. In reproducing figure 2, I will use their method of using  $\sigma^+$  values in one region of the plot and Yukawa-Tsuno parameters mixing  $\sigma$  and  $\sigma^-$  in the other region. When reproducing figures 3 and 4 of the paper, I will repeat the authors approach of creating new substituent constants using the Ingold-Yukawa-Tsuno equation with  $\sigma^n$ ,  $\Delta\sigma_R^+$  and  $\Delta\sigma_R^-$  parameters. Can I obtain the same results using the published data?

### Collect the Data

It is always a useful exercise to plot the data in a paper and compare to the plots that were presented by the authors. You can best understand great artists by imitating them at first. The authors did not document in detail how they determined the values of  $k_{A,R}$  values at pH 1.0, 3.8 and 11.5 that were used in the Hammett plots in the paper.<sup>6</sup> We will take these values as is.

The data from table 1 of the paper<sup>1</sup> is reproduced in table 1 below. We now have the data that was used to create the plots presented in the paper. Can we match what has been published? I will follow the methods of the authors, and attempt to reproduce the plots.

|           | Substituent        | $k_{A,R_{pH\ 11.5}}$<br>/ $10^{-3}s^{-1}$ | $k_{A,R_{pH\ 3.8}}$<br>/ $10^{-6}s^{-1}$ | $k_{A,R_{pH\ 1.0}}$<br>/ $10^{-6}s^{-1}$ | $pK_a$ |
|-----------|--------------------|-------------------------------------------|------------------------------------------|------------------------------------------|--------|
| <b>1c</b> | p-OCH <sub>3</sub> | 3.95                                      | 12.4                                     | 422                                      | -0.15  |
| <b>1e</b> | p-CH <sub>3</sub>  | 4.67                                      | 11.9                                     | 371                                      | -0.13  |
| <b>1d</b> | m-CH <sub>3</sub>  | 6.41                                      | 9.18                                     | 316                                      | -0.090 |
| <b>1b</b> | H                  | 3.68                                      | 7.94                                     | 286                                      | -0.090 |
| <b>1i</b> | p-Br               | 2.00                                      | 3.44                                     | 105                                      | -0.025 |
| <b>1g</b> | p-Cl               | 1.61                                      | 3.99                                     | 119                                      | -0.033 |
| <b>1f</b> | m-Cl               | 2.69                                      | 2.71                                     | 78.3                                     | 0.011  |
| <b>1h</b> | m-Br               | 31.2                                      | 2.64                                     | 74.6                                     | 0.007  |
| <b>1j</b> | p-CN               | 321                                       | 0.950                                    | 22.2                                     | 0.055  |
| <b>1k</b> | m-NO <sub>2</sub>  | 158                                       | 0.995                                    | 24.3                                     | 0.061  |
| <b>1l</b> | p-NO <sub>2</sub>  | 1190                                      | 0.610                                    | 12.4                                     |        |

### Replotting Figure 2

In figure 2 of the paper,<sup>1</sup> reproduced in figure 8 on the following page, is presented a Hammett plot for the observed rate constant  $k_{A,R_{pH\ 11.5}}$ . This is the rate constant observe at  $pH = 11.5$  and is dominated by the hydroxide-catalyzed reaction. The authors present a Yukawa-Tsuno plot using familiar  $\sigma$  and  $\sigma^-$  substituent parameters one region of the plot and a Brown-Okamoto plot with  $\sigma^+$  values in another. This plot features two distinct

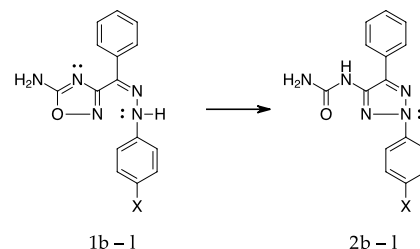


Figure 7: Rearrangement of (Z)-Arylhydrazones of 5-Amino-3-benzoyl-1,2,4-oxadiazole

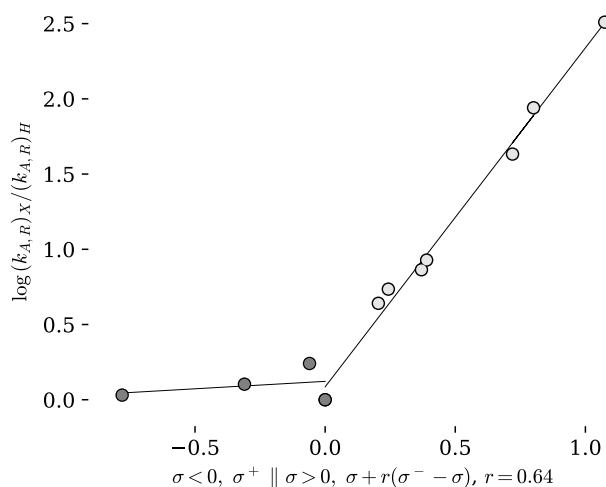
<sup>6</sup> I will presume that they used the observed rate and calculated the  $k_{A,R}$  value using the rate law corresponding to the mechanism that dominates at the given pH value.

Table 1: Data from table 1 of the paper. Observed overall rate constant values ( $k_{A,R_{pH}}$ ) are reported at pH 1.0, 3.8 and 11.5.  $pK_a$  is  $-\log 1/K$

slopes, indicating a change in reaction mechanism across the series of substituted reactants. The more electron-donating substituents show a negative slope of moderate magnitude while the electron-withdrawing substituents show a positive slope of greater magnitude.

The familiar Yukawa-Tsuno plot, with  $\sigma$  and  $\sigma^-$  values for the electron-withdrawing series, is appropriate for aniline-like nucleophiles. The use of a  $\sigma^+$  values for the handful of points in the electron-donating series seems dubious to me. The aniline nitrogen of the hydrazone group at the reaction centre is electron-rich and there is no empty space there to accept electron donation by resonance. I believe that this is a case of “it fits better” trumping theoretical rigour. There is no good reason to use  $\sigma^+$  values as the substituent constants for aniline-like nucleophiles.

First let us present our reproduction of figure 2 using the authors method. This plot is shown in figure 9.



We can see that there is a discrepancy between what the data gives us and the authors plot. The electron-withdrawing region ( $\sigma > 0$ ) shows that the Yukawa-Tsuno plots are very similar. Small differences can be attributed to slight variations between tables of substituent parameters found in the literature. However, the electron-donating region ( $\sigma < 0$ ) is very different. It almost looks like the data points for **1c**, **1d** and **1e** are reversed between the data in table 1 of the paper (reproduced in table 1 on the previous page) and the plot of figure 2 of the paper. I tend to trust the plots more than data tables that are prone to being altered in the typesetting process, but I can't say for sure.

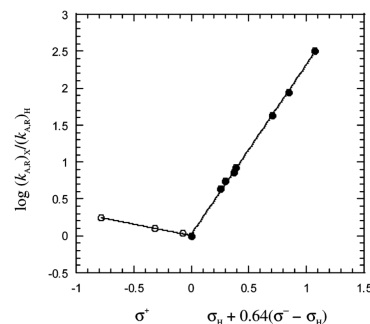


FIGURE 2. Plot of  $\log(k_{A,R})_X/(k_{A,R})_H$  at 313.1 K and at pS<sup>+</sup> 11.50 (see eqs 2 and 3) for the rearrangement of **1b**–**1** into the relevant triazoles **2b**–**1** versus  $\sigma^+$  or  $(\sigma_H + r^-(\sigma^- - \sigma_H))$  (YT plots).

Figure 8: Figure 2 from the paper<sup>1</sup>

Figure 9: A replot of figure 2 in the paper using the data in table 1 of the paper according to the equation in that figure. The open circles are plotted according to the Yukawa-Tsuno method with  $r = 0.64$  and my line fits shows that  $\rho = 2.25 \pm 0.09$  (very close to the authors value of  $\rho = 2.3$ ). The closed circles are plotted using Brown-Okamoto  $\sigma^+$  parameters and this portion of the plot deviates greatly from the authors version. For what its worth, my slope is  $\rho^+ = 0.12 \pm 0.09$  compared to the authors value of  $\rho^+ = -0.30$ .

Code for this plot can be explored via [Google Colab](#)

Table 2: Hammett substituent constants used in the attempt to reproduce figure 2 from the paper.

|                    | $\sigma$ | $\sigma^+$ | $\sigma^-$ |
|--------------------|----------|------------|------------|
| p-OCH <sub>3</sub> | -0.27    | -0.78      |            |
| m-OCH <sub>3</sub> | 0.11     |            |            |
| p-CH <sub>3</sub>  | -0.17    | -0.31      |            |
| m-CH <sub>3</sub>  | -0.07    |            |            |
| H                  | 0        |            |            |
| p-Br               | 0.23     | 0.15       |            |
| p-Cl               | 0.23     | 0.11       |            |
| m-Cl               | 0.37     |            |            |
| m-Br               | 0.39     |            |            |
| p-CN               | 0.66     |            | 1.00       |
| m-NO <sub>2</sub>  | 0.71     |            |            |
| p-NO <sub>2</sub>  | 0.78     |            | 1.27       |

### Conclusions for Figure 2

I will go no farther here. There is no point in trying to improve this plot while the mystery remains as to whether I should use the data as plotted or as written.<sup>7</sup> In the following section of this exploration I will obtain my own values for the rate constants of each pathway using curve fits of the pH-rate profiles. I will analyze the data for the hydroxide-catalyzed pathway there and present my own version of that plot when I have greater confidence in the data that I am using.

### Reproducing Figures 3 and 4 – The Ingold-Yukawa-Tsuno Equation

In the Hammett plots for  $k_{A,R_{pH} 3.8}$  and  $k_{A,R_{pH} 1.0}$ , the authors use a completely different method for constructing the substituent constant. The Ingold-Yukawa-Tsuno (IYT) equation combines the polar effect of the substituent with the resonance effect. The IYT equation is shown in eq. 9

$$\sigma' = \sigma^n + r^+ \Delta\sigma_R^+ + r^- \Delta\sigma_R^- \quad (9)$$

This equation will create a new substituent constant,  $\sigma'$  that takes the polar substituent constant  $\sigma^n$  and mixes in some amount of resonance donation or acceptance according to the proportionality factors  $r^+$  and  $r^-$ . Before we begin we should examine the parameters that the authors chose for the polar and resonance parameters. Where do  $\sigma^n$ ,  $\Delta\sigma_R^+$  and  $\Delta\sigma_R^-$  come from?<sup>8</sup>

### Wepster's $\sigma^n$ values

In the IYT approach used in the paper, *meta* substituents are considered to have only a polar effect. While *para* substituents will have an additional resonance contribution. Even the plain Hammett  $\sigma$  value for *para* substituents includes a resonance effect. For example, the methoxy group has an electron-withdrawing polar effect and an electron-donating resonance effect. The  $\sigma_m$  value for the methoxy group is +0.11, indicating that it is a polar electron-withdrawing group and yet the  $\sigma_p$  value is -0.27. This negative  $\sigma$  value demonstrates the resonance effect for *para* substituents. Can we isolate the two effects from each other?

The IYT equation presented by the authors uses  $\sigma^n$  values popularized by Wepster in the 1970's as the polar substituent constant for *para* groups,<sup>9,10</sup> These are often referred to as "normal"  $\sigma$  values and were determined using reactions where the reaction centre is spaced away from the phenyl ring by one or more alkyl groups. The hope is to insulate against any resonance connection between the ring and the reaction centre. There will still be a resonance effect included in the  $\sigma^n$  value because the phenyl ring carbon *para* to the substituent will be influenced by resonance effects but the reaction centre will only feel the polar effect of electron density at that carbon.<sup>11</sup>

<sup>7</sup> Spoiler-alert! You will soon see in part X that I am able to determine that the discrepancy is the result of a mix-up of data in the table. If only the data, data analysis and plots were documented using *Python* notebooks then these problems would be easily corrected. However, all we can do is guess at which data is correct – the plots or the tables. Or perhaps neither?

<sup>8</sup> Take note that the  $\sigma^n$  values used here are not Hammett's  $\sigma$  values. They are Wepster's values. *Vide infra*.

<sup>9</sup> "Substituent effects. IV. Reexamination of  $\sigma^n$ ,  $\Delta\sigma_R^+$  and  $\Delta\sigma_R^-$  values. Arylacetic acids and other insulated systems." A.J. Hoefnagel, B.M. Wepster, *J. Am. Chem. Soc.*, 1973, 95, 5357–5366. <https://doi.org/10.1021/ja00797a041>

<sup>10</sup> "Substituent effects. III. Dissociation constants of  $\beta$ -arylpropionic acids,  $\beta$ -arylisovaleric acids, N-arylglycines, aryloxyacetic acids, N-aryl- $\beta$ -alanines, and some related systems." A.J. Hoefnagel, J.C. Monshouwer, E.C.G. Snorn, B.M. Wepster *J. Am. Chem. Soc.*, 1973, 95, 5350–5356 <https://doi.org/10.1021/ja00797a040>

<sup>11</sup> This is not exactly true. The phenyl ring is a large swirling donut of resonance and the resonance effect of the ring can be transmitted through space if orbitals are aligned.



The  $\sigma^n$  set of substituent constants is rarely used today. In fact, this paper is the first time that I have ever encountered them. In modern times the  $\sigma_I$  polar substituent parameter is used to represent the inductive effect of *para* substituents and better isolates the inductive effect from the resonance contribution of a substituent. The corresponding author (Spinelli) has been publishing work on this reaction since 1976<sup>12</sup> and began using the IYT equation in a paper published in 1978, in which they published a plot that looks almost exactly like the plots in our paper here.<sup>13</sup> Should these old dogs learn some new tricks?

### Wepster's $\Delta\sigma_R^+$ and $\Delta\sigma_R^-$ Values

In Wepster's system, each substituent constant has two components, the normal ( $\sigma^n$ ) and the resonance component,  $\Delta\sigma_R^+$  or  $\Delta\sigma_R^-$ . These resonance components were calculated from Brown-Okamoto values as  $\Delta\sigma_R^+ = \sigma^+ - \sigma^n$  and  $\Delta\sigma_R^- = \sigma^- - \sigma^n$ . The values for  $\sigma^n$  and the calculated values for  $\Delta\sigma_R^+$  and  $\Delta\sigma_R^-$  are presented in table 3,

|    | Substituent        | $\sigma$ | $\sigma^+$ | $\sigma^-$ | $\sigma^n$ | $\Delta\sigma_R^+$ | $\Delta\sigma_R^-$ |
|----|--------------------|----------|------------|------------|------------|--------------------|--------------------|
| 1c | p-OCH <sub>3</sub> | -0.27    | -0.78      |            | -0.09      | -0.69              |                    |
|    | m-OCH <sub>3</sub> | 0.11     |            |            |            |                    |                    |
| 1e | p-CH <sub>3</sub>  | -0.17    | -0.31      |            | -0.10      | -0.21              |                    |
| 1d | m-CH <sub>3</sub>  | -0.07    |            |            |            |                    |                    |
| 1b | H                  | 0        |            |            | 0          |                    |                    |
| 1g | p-Cl               | 0.23     | 0.11       |            | 0.29       | -0.18              |                    |
| 1f | m-Cl               | 0.37     |            |            |            |                    |                    |
| 1i | p-Br               | 0.23     | 0.15       |            | 0.30       | -0.15              |                    |
| 1h | m-Br               | 0.39     |            |            |            |                    |                    |
| 1j | p-CN               | 0.66     |            | 1.00       | 0.70       |                    | 0.30               |
| 1k | m-NO <sub>2</sub>  | 0.71     |            |            |            |                    |                    |
| 1l | p-NO <sub>2</sub>  | 0.78     |            | 1.27       | 0.78       |                    | 0.49               |

<sup>12</sup> "Mononuclear heterocyclic rearrangement. Note I. Kinetic study of the rearrangement of the phenylhydrazones of 3-benzoyl-5-phenyl-1,2,4-oxadiazole into 2,5-diphenyl-4-benzoylamino-1,2,3-triazole." D. Spinelli, A. Corrao, et al. *J. Heterocyclic Chem.*, 1976, 13, 357-360. <https://doi.org/10.1002/jhet.5570130229>

<sup>13</sup> "Mononuclear heterocyclic rearrangements. Part 2. Substituent effects on the rate of rearrangement of some arylhydrazones of 3-benzoyl-5-phenyl-1,2,4-oxadiazole into 2-aryl-4-benzoylamino-5-phenyl-1,2,3-triazole, at pS<sup>+</sup> 3.80", D. Spinelli, V. Frenna, et al., *J. Chem. Soc., Perkin Trans.*, 1978, 2 19-22. <https://doi.org/10.1039/P29780000019>

Table 3: The parameter set used by the authors. From table 3 in reference 13. The m-OCH<sub>3</sub> substituent constant value is presented for comparison purposes and is not part of the experimental series.

### Reproducing Figures 3 and 4 – Hammett Plots for $k_{A,RpH\ 3.8}$ and $k_{A,RpH\ 1.0}$

The Hammett plots for  $k_{A,RpH\ 3.8}$  and  $k_{A,RpH\ 1.0}$  using the IYT equation are presented in figures 10 and 12 on the next page. They seem identical to the corresponding figure 3 and 4 of the paper<sup>1</sup> as shown in the margin as figures 11 and 13.

The values of  $r^+$  and  $r^-$  are varied until the best fit is found. In the paper, the authors have found the best fit when  $r^+ = 0.10$  and  $r^- = 0.24$ .<sup>14</sup> If that is the case, then the  $\sigma'$  value for the p-OCH<sub>3</sub> substituent is -0.16.<sup>15</sup> Take note that this is a smaller value than the  $\sigma_p = -0.27$  substituent constant. This seems to indicate that the resonance effect in plain Hammett  $\sigma$  values is reduced in this case. Perhaps the electron-rich aniline nitrogen of the hydrazone repels the electron-donating resonance effect a little bit?<sup>16</sup>

<sup>14</sup> These values indicate a weak resonance connection with the reaction centre, which begs the question as to why we would use a multiparameter approach when perhaps a plain Hammett plots using only  $\sigma$  values might suffice. I will address this idea later in this exploration.

<sup>15</sup>  $-0.09 + 0.10(-0.69) + 0.24(0) = -0.16$

<sup>16</sup> Or perhaps we should not bother with such a minor contribution and use the more common form of the Yukawa-Tsuno equation, mixing only  $\sigma$  and  $\sigma^-$ . I will present more on this idea later.

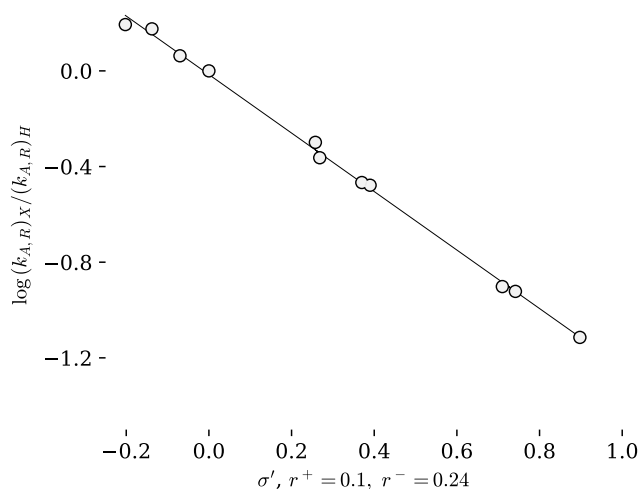


Figure 10: A replot of figure 3 in the paper using the data in table 1 on page 6 and table 3 on the previous page.

Code for this plot can be explored via [Google Colab](#)

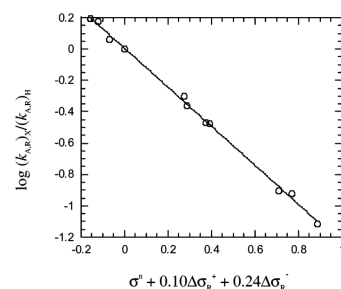


FIGURE 3. Plot of  $\log(k_{A,R})_X / (k_{A,R})_H$  at 313.1 K and at  $pS^+ 3.80$  (see eq 4) for the rearrangement of **1b-1** into the relevant triazoles **2b-1** versus  $(\sigma^0 + r^+\Delta\sigma^+ + r^-\Delta\sigma^-)$  ("overall" IYT plot).

Figure 11: Figure 3 from the paper<sup>1</sup>

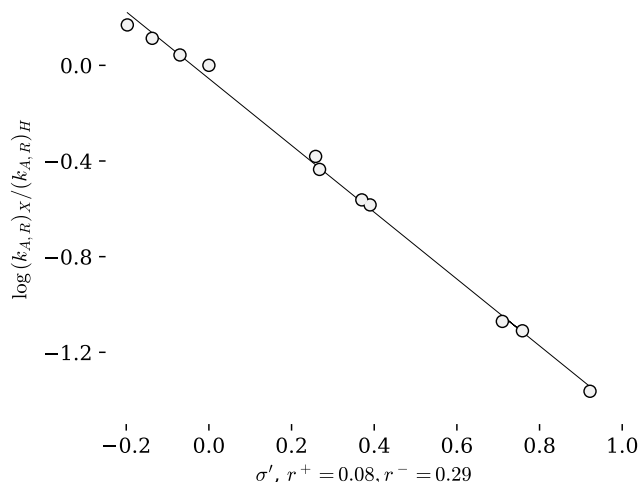


Figure 12: A replot of figure 4 in the paper using the data in table 1 on page 6 and table 3 on the previous page.

Code for this plot can be explored via [Google Colab](#)

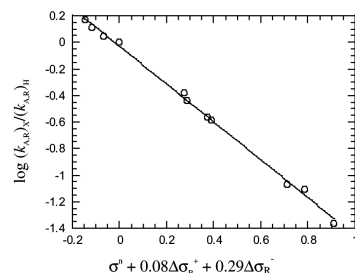


FIGURE 4. Plot of  $\log(k_{A,R})_X / (k_{A,R})_H$  at 313.1 K and at  $pS^+ 1.00$  (see eq 10) for the rearrangement of **1b-1** into the relevant triazoles **2b-1** versus  $(\sigma^0 + r^+\Delta\sigma^+ + r^-\Delta\sigma^-)$  ("overall" IYT plot).

Figure 13: Figure 4 from the paper<sup>1</sup>

### Conclusions for Figures 3 and 4

The reaction constants,  $\rho$  in both mechanisms are similar. For the substituent effect on  $k_{A,R_{pH\ 3.8}}$  we observe that  $\rho = -1.22 \pm 0.02$ , and for  $k_{A,R_{pH\ 1.0}}$  we observe that  $\rho = -1.40 \pm 0.03$ . this compares well to the values in the paper of  $-1.24 \pm 0.02$  and  $-1.43 \pm 0.02$ , respectively. However, I find the IYT approach to be a very niche idea. Later in this exploration, we will use the standard Yukawa-Tsuno approach.

### Part 3: Analyzing the Data Ourselves

In this section, I will obtain my own values for the rate constants of each mechanism by performing curve fits of the pH-rate data against the model described by eq. 7 on page 4. The pH-rate profile for reactant **1b** was determined previously at a different temperature.<sup>17</sup> We will show how the data for **1b** was adjusted so that it could be included in this data set.

Then we will curve fit all the pH-rate profiles and create our own table of rate constants. Ultimately, we will obtain the rate constants for each of the mechanisms that compose our model. Will have values for  $k_{\text{OH}}$ ,  $k_{\text{H}_2\text{O}}$  and  $k_{\text{H}}/K_{\text{a}}$ . These relate to the authors' values of  $k_{\text{A,R}_{\text{pH}} 11.5}$ ,  $k_{\text{A,R}_{\text{pH}} 3.8}$  and  $k_{\text{A,R}_{\text{pH}} 1.0}$ , which are the observed rate constants at each give pH value.

#### Big Data Tables

The first step is to transcribe the data tables found in the supplementary material. The paper has ten data sets of  $k_{(\text{A,R})}$  values (the authors' term for the observed rate constant,  $k_{\text{obs}}$ , of the rearrangement) for substituted arylhydrazones but does not have data for the unsubstituted arylhydrazone (**1b**, X = H). That data is presented in another earlier report<sup>17</sup> but the data is given for a different temperature.

#### Temperature Correction for the Unsubstituted Case, **1b**

To obtain a data set for **1b**, we must recalculate the data that was reported at 293 K<sup>17</sup> to get values at 313 K. The data set for **1b** includes activation parameters,  $\Delta H^\ddagger$  and  $\Delta S^\ddagger$ , as well as the rates constants,  $k_{(\text{A,R})293\text{K}}$ , reported at 293 K.<sup>17</sup> Using the Eyring equation, I calculated the corresponding rate constants,  $k_{(\text{A,R})313\text{K}}$ , at 313 K.

Figure 14 presents the values of  $k_{(\text{A,R})313\text{K}}$  that were calculated by using activation parameters plotted against the measured values of  $k_{(\text{A,R})293\text{K}}$ . The slope of the line is set to 1.0 and the intercept will be the average magnitude difference between the two sets of data. The values calculated at 313 K were on average 10.66-fold greater than the observed values at 293 K.

Figure 15 on the next page presents the values of  $k_{(\text{A,R})313\text{K}}$  calculated by using activation parameters and the measured values of  $k_{(\text{A,R})293\text{K}} \times 10.66$ . Both are plotted together against pH. They track perfectly together. The measured values have higher precision, so I will be using the data set derived from them combined with the factor of 10.66 to adjust for temperature. Feel free to use the calculated Eyring values yourself and compare how that change affects this analysis – the code is available to inspect and use via [Github](#) and [Google Colab](#).

<sup>17</sup> “The First Kinetic Evidence for Acid Catalysis in a Monocyclic Rearrangement of Heterocycles: Conversion of the Z-Phenylhydrazone of 5-Amino-3-benzoyl-1,2,4-oxadiazole into N,5-Diphenyl-2H-1,2,3-triazol-4-ylurea.” B. Cosimelli, V. Frenna, S. Guernelli, C.Z. Lanza, G. Macaluso, G. Petrillo, D. Spinelli, *J. Org. Chem.*, 2002, 67, 8010-8018. <https://doi.org/10.1021/jo026039z>

The Eyring equation

$$k_{\text{obs}} = \kappa \frac{k_{\text{B}}}{h} T e^{\frac{\Delta S^\ddagger}{T}} e^{\frac{\Delta H^\ddagger}{RT}}$$

$$\kappa = 1$$

$$h = 6.63 \times 10^{-34} \text{ J s}$$

$$k_{\text{B}} = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

$$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$$

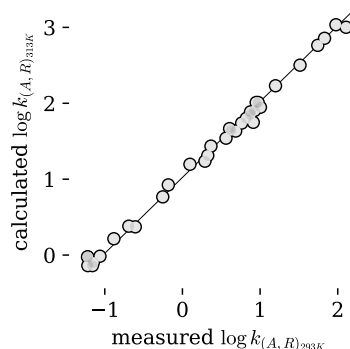


Figure 14: Correlation of calculated and experimental rate constants for reactant **1b**. Calculated values are determined using the Eyring equation and the  $\Delta S^\ddagger$  and  $\Delta H^\ddagger$  values in the referenced paper.<sup>17</sup>

Code for this plot can be explored via [Google Colab](#)

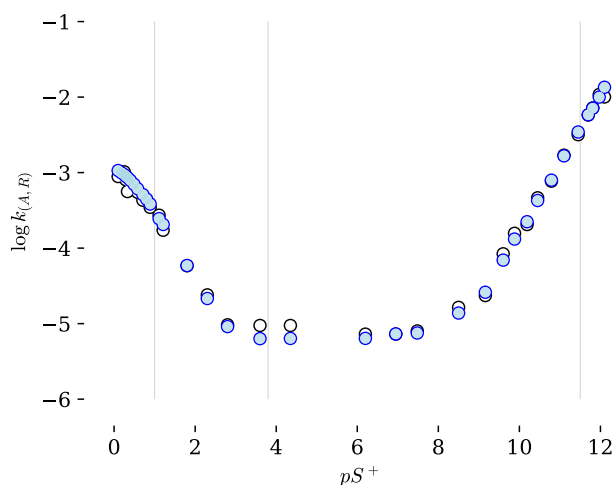


Figure 15: Plot of pH-rate profiles for measured and calculated rate constants at 313K. Blue points are measured  $\times 10.66$ ; black points are calculated using the Eyring Equation.

Code for this plot can be explored via [Google Colab](#)

### Curve Fitting a pH-Rate Profile

I will use the model from eq. 7 on page 4 to obtain best fit parameters for the variables. At the end of this section we will have developed our own method for determining values for  $pK_a$ ,  $k_H$ ,  $k_{H_2O}$  and  $k_{OH}$ .

An example is shown in figure 16 where I used the CurveFit function in the SciPy package of *Python* and curve fit the data against the model.

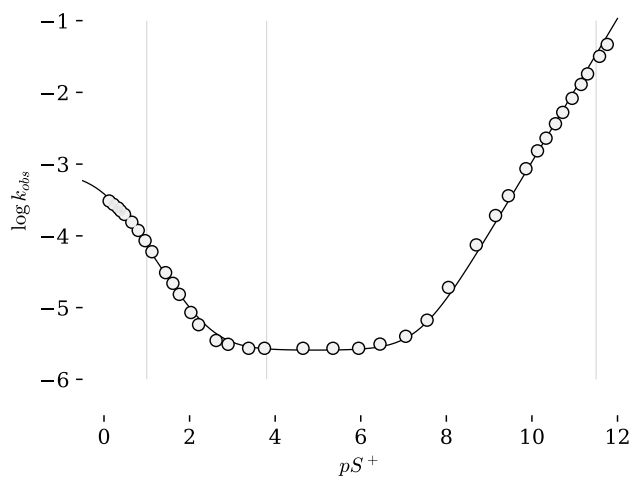


Figure 16: pH-rate profile for rearrangement of 1f in 1:1 dioxane/water. Vertical lines highlight the values of  $pS^+ = 1.0, 3.8$  and  $11.5$ .

Values determined were...

$$\begin{aligned} k_{OH} &= 10.2 \pm 0.3 \text{ M}^{-1} \text{ s}^{-1} \\ k_{H_2O} &= 2.54 \pm 0.14 \times 10^{-6} \text{ s}^{-1} \\ k_H &= 7.75 \pm 0.18 \times 10^{-4} \text{ s}^{-1} \\ pK_{a_{AH}} &= 0.01 \pm 0.13 \end{aligned}$$

The values reported in the paper were...

$$\begin{aligned} k_{OH} &= 8.5 \text{ M}^{-1} \text{ s}^{-1} \\ k_{H_2O} &= 2.71 \times 10^{-6} \text{ s}^{-1} \\ k_H &= 7.83 \times 10^{-4} \text{ s}^{-1} \\ pK_{a_{AH}} &= -0.01 \end{aligned}$$

Code for this plot can be explored via [Google Colab](#)

The results from the curve fit and the values reported by the authors are collected in the caption of figure 16. the values from my own curve fits are very close to those reported by the authors. However I do observe some systematic deviation between the data and the model in the residual plot shown in figure 17.

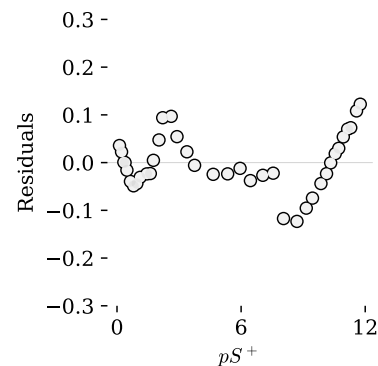


Figure 17: Residual plot for curve fit of pH-rate profile presented in figure 16

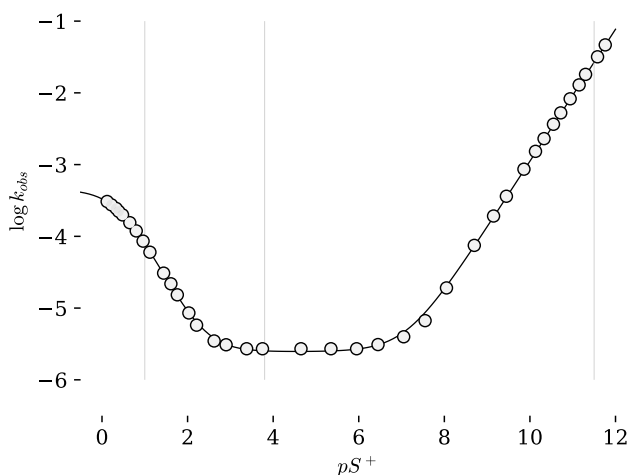
### Into the Rabbit Hole: What's up with that Slope?

Examine the residual plot in figure 17 on the previous page. The deviations at the extremes of the residual plot are due to the slopes of the lines not aligning with the model. The authors allow the slope of  $\approx 0.9$  in their curve fit as they are performing a simple linear regression of the “linear portion” of the data.

My slope is 1.0 (by definition) due to the mathematical model that I am using. The observed slopes of  $\approx 0.9$  observed in the data are real and cannot be explained by my model. This implies that the reaction order on  $[\text{OH}^-]$  is 0.9 and the rate law expressed in eq. 1 on page 3 would then be  $d/dt[\text{P}] = k_{\text{OH}}[\text{OH}^-]^{0.9}[\text{A}]$ . Likewise, the slope for linear region of the acid-catalyzed reaction is slightly greater than 1.0, although the authors report it as being about 0.9 as well.<sup>1</sup> A better explanation would be the  $\text{pS}^+$  acidity function is not quite a perfect description of the activity of proton in their system.<sup>18</sup>

Figures 18 and 19 present the modified model that includes the non-unitary reaction orders for  $[\text{H}^+]$  and  $[\text{OH}^-]$ . The model used to fit the data is presented in eq. 10.

$$k_{\text{obs}} = k_{\text{OH}} \frac{K_{\text{W}}}{[\text{H}^+]^{0.93}} + k_{\text{H}_2\text{O}} + k_{\text{H}} \frac{[\text{H}^+]^{1.10}}{K_{\text{a}} + [\text{H}^+]^{1.10}} \quad (10)$$



The residuals presented in figure 19 show a much better fit in the regions that describe  $[\text{H}^+]$  and  $[\text{OH}^-]$ . The two deviations are where the curve of the model can't follow the tighter curve in the data as we transition between the three regions of the plot. There is clearly something about the physical chemistry of acid/base catalysis in aqueous/organic solvent mixtures that I do not understand. The issue is that the authors do not address this either.

<sup>18</sup> The best fit for the data is achieved when I used a reaction order of 1.10 for  $[\text{H}^+]$  and 0.93 for  $[\text{OH}^-]$ . There is no theoretical reason to do this that, as far as I know. The fact that the added parameters improved the fit doesn't necessarily mean anything and we should never add parameters without a real reason to do so.

Figure 18: pH-rate profile for rearrangement of 1f in 1:1 dioxane/water. A model that included hypothetical non-unitary reaction orders (see eq. 10) was used.

Values determined were...

$$k_{\text{OH}} = 5.2 \pm 0.2 \text{ M}^{-1} \text{ s}^{-1}$$

$$k_{\text{H}_2\text{O}} = 2.43 \pm 0.07 \times 10^{-6} \text{ s}^{-1}$$

$$k_{\text{H}} = 5.06 \pm 0.39 \times 10^{-4} \text{ s}^{-1}$$

$$\text{p}K_{\text{aAH}} = 0.33 \pm 0.04$$

The values reported in the paper were...

$$k_{\text{OH}} = 8.5 \text{ M}^{-1} \text{ s}^{-1}$$

$$k_{\text{H}_2\text{O}} = 2.71 \times 10^{-6} \text{ s}^{-1}$$

$$k_{\text{H}} = 7.83 \times 10^{-4} \text{ s}^{-1}$$

$$\text{p}K_{\text{aAH}} = -0.01$$

Code for this plot can be explored via [Google Colab](#)

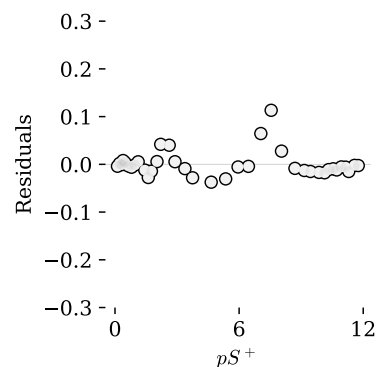


Figure 19: Residual plot for curve fit of pH-rate profile presented in figure 18

### *There can be Only One*

Slopes of acid and base catalysis in pH-rate profiles have a slope of one, by definition. The non-unitary slopes for the pH-rate profile for acid catalysis and base catalysis need to be explained. The authors have a set of references (see reference 11 in the paper<sup>1</sup>) for technical background information on acid catalysis. They reference a whole bookshelf of classic textbooks in physical and physical organic chemistry when they mention the slope and specifically refer the reader to a textbook by Keith Laidler, an important Canadian chemist (ref 11a in the paper.) They do not discuss the matter further and neither does Laidler's textbook.<sup>19,20</sup> The author of the paper has not read that work – or they are hoping that a throwaway reference is all they need to convince the lazy reader that no one needs to look into this unusual reaction order.

The model presented in eq. 10 on the previous page with its non-unitary reaction orders for  $[H^+]$  and  $[OH^-]$  does not have a basis in theory (that I am aware of) and is best understood as the authors using empirical “fudge factors” to improve the fit. By leaving this matter unaddressed, the authors are revealing that they have no answer and would prefer that you ignore it too. Or, perhaps the answer is so obvious that they assume the reader doesn't need any help. Am I the problem – or does the  $pS^+$  scale need to be tweaked? You be the judge.

### *Choosing the Model*

Which model should we use? We have created a mathematical model for the pH-rate profile as shown in eq. 7 on page 4. This model was fundamentally the same as that presented in equation 1 in the paper.<sup>1</sup> The authors report non-unitary reaction orders in their acid and base-catalyzed reactions. They offered no explanation and referenced a source that did not address this observation. My own analysis, presented in figure 16 on page 12 and figure 18 on the previous page, showed that the slopes were approximately -1.1 and 0.9 and this was used to define the model described in eq. 10 on the previous page. My model cannot account for these non-unitary slopes and I can offer no explanation myself except to suggest that the  $pS^+$  scale used by the authors is the source of a systematic error.

The small error due to the imperfect fit of the model should not affect the analysis of the results. In the next section we will reproduce and comment upon the Hammett plots for the rates of reaction in each of the three regions of the pH-rate profile. We will use eq. 7 as the model, declaring firmly that the kinetic order for proton concentration is unitary, as nature intended.

<sup>19</sup> I was able to get my hands of the 3<sup>rd</sup> edition from 1987 (its on a shelf in the library – QD501.L17) and not the 2<sup>nd</sup> edition from 1965 that was referenced by the authors. (It is also available from the library, but it is in compact storage in the mouldy basement. I'm not going there without a hazmat suit.)

<sup>20</sup> Wait a minute – this paper was written in 2006. Why are they referencing the edition from 40 years earlier when more recent editions are available?

### Curve Fitting All the Data

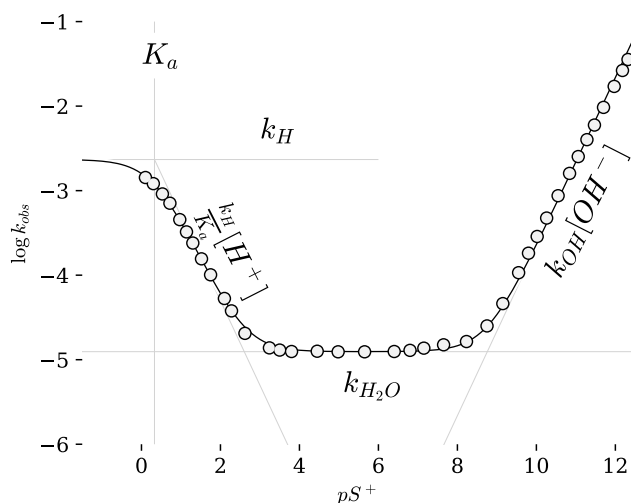
We now have a set of eleven pH-rate profiles for the rearrangements of various substituted arylhydrazones, **1b** – **1l**. Ten are from the paper that we are exploring,<sup>1</sup> and the data for **1b** was found in another paper<sup>17</sup> and adjusted for temperature. The data sets were all transcribed and collected in csv text files as described in table 4.

In the first section above we discussed the model for the observed rate constant for rearrangement,  $k_{(A,R)}$ . It is the total rate constant for the three reactions involved. I prefer to use the term  $k_{obs}$ . The model is presented in eq. 7 on page 4 and is reproduced below.

$$k_{obs} = k_{OH} \frac{K_W}{[H^+]} + k_{H_2O} + k_H \frac{[H^+]}{K_a + [H^+]}$$

I then wrote some code to perform curve fit using the above model to obtain the best values for  $k_{OH}$ ,  $k_{H_2O}$ ,  $k_H$  and  $K_a$ . The code is available to inspect and use via [Github](#) and [Google Colab](#).

A example plot with the curve fit line is presented in figure 20. You will observe that the value of  $K_a$  for the protonated reactant is defined by very little information. If the  $K_a$  cannot be accurately determined, neither can the  $k_H$  value. In the region where  $K_a > [H^+]$ , we see that the rate constant is dominated by the term  $\frac{k_H}{K_a} [H^+]$ . So the value of  $k_H/K_a$  will give us the best value for the rate constant in the acid-catalyzed region of the pH-rate profile. If the curve fit overvalues  $K_a$ , it will undervalue  $k_H$  by the same factor and cancel out any errors.



We will use the  $k_H/K_a$  value to represent the rate constant for the specific acid reaction (dominant at  $pH = 1.0$ ); the  $k_{H_2O}$  value for uncatalyzed rate ( $pH = 3.8$ ); and the  $k_{OH}$  value general base reaction ( $pH = 11.5$ ).

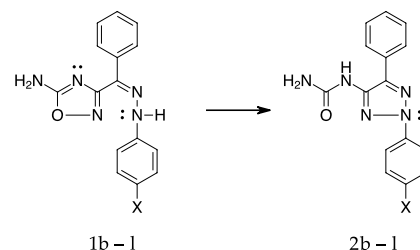


Table 4: Data files created from the supplementary data of the paper.<sup>1</sup> These will be used in the curve fit analysis.

|           | Substituent        | File Name   |
|-----------|--------------------|-------------|
| <b>1b</b> | H*                 | 1b-data.csv |
| <b>1c</b> | p-OCH <sub>3</sub> | 1c-data.csv |
| <b>1d</b> | m-CH <sub>3</sub>  | 1d-data.csv |
| <b>1e</b> | p-CH <sub>3</sub>  | 1e-data.csv |
| <b>1f</b> | m-Cl               | 1f-data.csv |
| <b>1g</b> | p-Cl               | 1g-data.csv |
| <b>1h</b> | m-Br               | 1h-data.csv |
| <b>1i</b> | p-Br               | 1i-data.csv |
| <b>1j</b> | p-CN               | 1j-data.csv |
| <b>1k</b> | m-NO <sub>2</sub>  | 1k-data.csv |
| <b>1l</b> | p-NO <sub>2</sub>  | 1l-data.csv |

All data are reported at 313 K

\*Data for **1b** obtained from a previous work<sup>17</sup> at 293 K. Rates were adjusted to 313 K as described for figures 14 and 15.

Figure 20: Plot of pH-rate profile for **1c**.

This plot, including all the annotations, was created using *Matplotlib* tools in *Python*.

Code for this plot can be explored via [Google Colab](#)

### Automated Curve Fitting and Plotting

Rather than analyze each data set individually, I wrote a program to use the code discussed above to curve fit and plot all eleven data sets automatically.<sup>21</sup> This code reads in each of the eleven data files in turn. With each case, the program performed a curve fit using the above model as described above and created a plot. The code is available to inspect and use via [Github](#) and [Google Colab](#).

The output is a data table of the kinetic parameters and the plots is saved in pdf format. The result data is written to a file named `results.csv`. This file will then be read in by the code for the various Hammett plots that are presented later in this section. The individual plots are presented in figure 37 on page 29 in the appendix at the end of this document.

<sup>21</sup> Always seek to automate your workflows. Get a computer to do the work that you are being paid to do while you get paid even more doing something else. AI will not steal your job, it will make you rich.

### The Results

We now have values that describe the rate constant for each of the three reactions and for the  $pK_a$  of the protonated reactant. The results from the curve fit to the model for the pH-rate profile are presented in table 5. The equivalent values based on the results in table 1 of the paper<sup>1</sup> are presented in table 1.

|    | Substituent        | $k_{OH}$<br>$/M^{-1}s^{-1}$ | $k_{H_2O}$<br>$/10^{-6}s^{-1}$ | $k_H/K_a$<br>$/10^{-3}M^{-1}s^{-1}$ | $pK_a$       |
|----|--------------------|-----------------------------|--------------------------------|-------------------------------------|--------------|
| 1c | p-OCH <sub>3</sub> | 2.180 ± 0.078               | 12.45 ± 0.41                   | 4.99 ± 0.26                         | 0.33 ± 0.08  |
| 1e | p-CH <sub>3</sub>  | 1.560 ± 0.056               | 11.97 ± 0.34                   | 4.35 ± 0.20                         | 0.34 ± 0.06  |
| 1d | m-CH <sub>3</sub>  | 1.37 ± 0.09                 | 8.78 ± 0.33                    | 3.50 ± 0.15                         | 0.23 ± 0.07  |
| 1b | H                  | 1.31 ± 0.10                 | 6.62 ± 0.35                    | 3.41 ± 0.22                         | 0.25 ± 0.09  |
| 1g | p-Cl               | 6.31 ± 0.23                 | 3.82 ± 0.23                    | 1.190 ± 0.084                       | 0.09 ± 0.15  |
| 1i | p-Br               | 7.73 ± 0.34                 | 3.29 ± 0.20                    | 0.986 ± 0.079                       | 0.01 ± 0.18  |
| 1f | m-Cl               | 10.73 ± 0.30                | 2.54 ± 0.14                    | 0.759 ± 0.050                       | -0.01 ± 0.15 |
| 1h | m-Br               | 11.64 ± 0.26                | 2.42 ± 0.12                    | 0.719 ± 0.049                       | 0.02 ± 0.14  |
| 1k | m-NO <sub>2</sub>  | 75.30 ± 0.63                | 0.937 ± 0.072                  | 0.220 ± 0.018                       | -0.13 ± 0.27 |
| 1j | p-CN               | 143.90 ± 0.37               | 0.825 ± 0.065                  | 0.204 ± 0.016                       | -0.04 ± 0.18 |
| 1l | p-NO <sub>2</sub>  | 635.2 ± 3.8                 | 0.471 ± 0.054                  | 0.109 ± 0.012                       | —            |

A plot of all the pH-rate profiles combined together is presented in figure 21 on the following page. You might be able to observe that the  $K_a$  value in the case of 1l cannot be determined and was set by the bounds of the curve fit. The individual plots in the appendix is available for your inspection as figure 37 on page 29.

Table 5: Curve fit parameters with standard deviations. Compare these values to those reported by the authors presented in table 1 on page 6



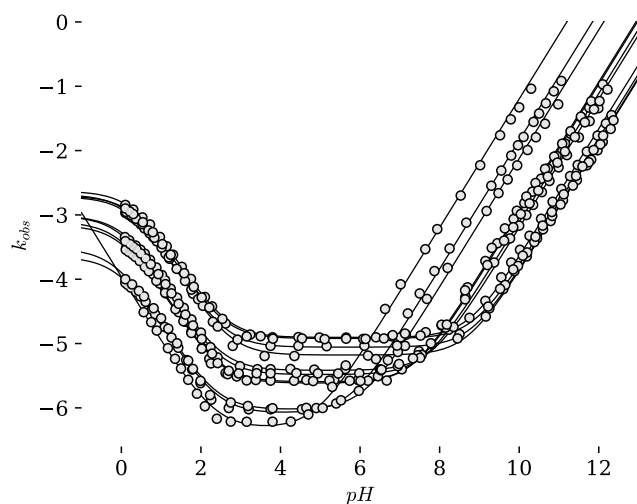


Figure 21: Plot of pH-rate profile for 1b-l combined in a single plot. The curve fit lines are included. The plots are presented individually in the appendix.

Take a moment to appreciate how much work was put into this paper. Each data point is the result of a separate experiment. Many of these experiments were performed at several temperatures so that the  $\Delta S^\ddagger$  and  $\Delta H^\ddagger$  values could be determined by the Eyring plot. The experiments in the hydroxide-catalyzed region were measured at multiple buffer concentrations and extrapolated to zero buffer concentration.

Code for this plot can be explored via [Google Colab](#)

#### Part 4: The Hammett Plots Again

We have determined the kinetic parameters for the rate law that describes pH-rate profiles for a set of eleven reactants in a Hammett series. We will now perform a Hammett plot analysis for each rate constant that contributes to the rate law and compare the results to the authors' work that was discussed above. I will also offer alternate methods of analysis that I feel would better stand up to scrutiny.

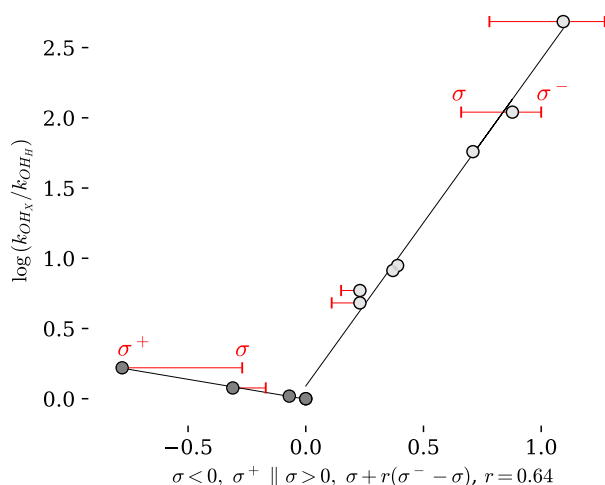
After curve fitting all eleven of these data sets with our model, we now have a data set of rate constants for each pathway calculated for all of the reactants. We can now repeat the Hammett analysis using this data and see if we get the same results.

#### Replotting Figure 2 Using Our Own Numbers

Recall the when we plotted the Hammett plot for the rate observed in the basic region ( $pH = 11.5$ ) we observed a plot that was significantly different than the published version. Compare figures 9 and 8 on page 7 again to remind yourself.

In figure 22 on the next page is presented the same plot again (using the same incorrect set of  $\sigma^+$  parameters for the electron-donating portion) but using the  $k_{OH}$  values determined from the curve fits and listed in table 5 on the previous page instead of the corresponding  $k_{A,R_{pH\ 11.5}}$  values reported by the authors and reproduced in table 1 on page 6.

As you can see, this plot is very similar to the plot in figure 2 of the paper.<sup>1</sup> Clearly the difference between my version of the authors' plot and the published version was a typographical error in the published data table. When I calculated the results myself using the raw data sets, I was able to reproduce



the plot accurately. That doesn't make it right, though.

The authors used Brown-Okamoto  $\sigma^+$  substituent parameters for the electron-donating substituents ( $\sigma < 0$ ). They did this for no other reason than an improvement in the observed line fit. The hydrazone group in the starting material, **1b-l**, has a lone pair that can donate electrons via resonance to an electron-acceptor, not an empty p-orbital that can accept electrons from a donor. The product of the rearrangement, **1b-l**, is also an aniline-like nitrogen with a lone pair. One would presume that the transition-state between the two also has little opportunity to be a resonance electron-acceptor.

### Replotting Figure 2 the Right Way

The Yukawa-Tsuno equation is a multi-parameter LFER approach that seeks to combine inductive Hammett  $\sigma$  parameters and the  $\sigma^+$  or  $\sigma^-$  resonance parameters. These resonance parameters are used for systems where there is a direct connection between a substituent and a reaction centre. Anilines are a classic example and we see an aniline-like group as the nucleophile in the rearrangement being considered here. Resonance electron-acceptors like cyano groups and nitro groups would have an outsized effect on electron density at the nitrogen due to the ability to withdraw electrons via resonance in addition to the overall electronic effect on the phenyl ring. Figure 23 displays an example.

I conclude that  $\sigma^+$  parameters should not be used and that  $\sigma^-$  parameters are appropriate for a hydrazone.<sup>22</sup> We do see significant resonance effect as demonstrated by the significant value for  $r$  in the Yukawa-Tsuno plot for the electron-withdrawing region of the plot. I decided to replot the data using an  $x$ -axis constructed with the Yukawa-Tsuno equation only. This will effectively result in using plain Hammett  $\sigma$  values on the electron-donating side,

Figure 22: A replot of figure 2 in the paper using  $k_{OH}$  values reported in table 5. The open circles are plotted according to the Yukawa-Tsuno method with  $r = 0.64$  and my line fits shows that  $\rho = 2.32 \pm 0.09$  (very close to the authors value of  $\rho = 2.3$ .) The closed circles are plotted using Brown-Okamoto  $\sigma^+$  parameters and this portion of the plot is now identical to the authors' version. My slope is  $\rho^+ = -0.28 \pm 0.01$  compared to the authors value of  $\rho^+ = -0.30$ .

The red lines denote the range between  $\sigma$  and  $\sigma^+$  or  $\sigma^-$ . You can see how the Yukawa-Tsuno plot "slides" the points between  $\sigma$  and  $\sigma^-$  to maximize the fitting score.

You can also see the result of using  $\sigma^+$  rather than  $\sigma$  values in the electron-donating region. This choice changed that section of the plot enormously. Is there any justification for using  $\sigma^+$  with aniline nucleophiles? No, there is not.

Code for this plot can be explored via [Google Colab](#)

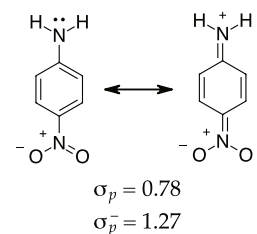
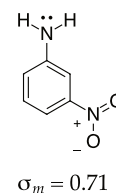


Figure 23: Examples of resonance contribution in  $\sigma^-$  parameters

<sup>22</sup> Just because I have a confident attitude and a winning smile does not mean that using the Yukawa-Tsuno plot is correct. Perhaps the authors' version is best. But, I feel that we should plot the data using a simple Yukawa-Tsuno plot and avoid the use of including a different set of  $\sigma^+$  parameters without a good reason other than that the three-point line fit looks better with  $\sigma^+$ . What do you think?

where substituents do not have  $\sigma^-$  values that differ from  $\sigma$  values. Figure 24 presents the plot. We see that the slope for the electron-withdrawing region is has a much greater magnitude (and the fit is not as good – spreading the points out with  $\sigma^+$  values would improve the fit, but there is no justification to do that.

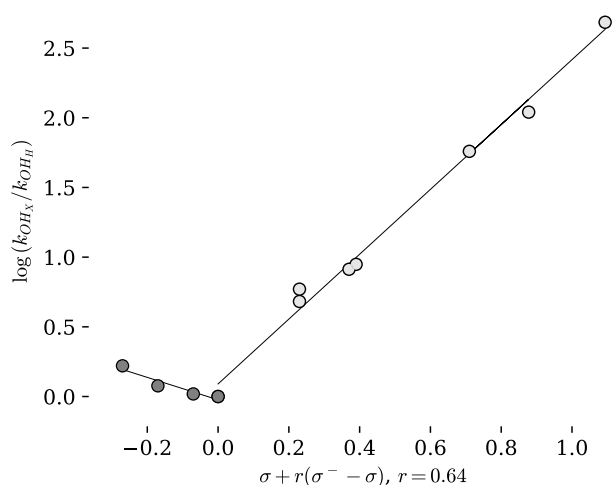


Figure 24: A replot of figure 2 in the paper using  $k_{OH}$  values reported in table 5. The open circles are plotted according to the Yukawa-Tsuno method with  $r = 0.64$  and my line fits shows that  $\rho = 2.32 \pm 0.09$  (very close to the authors value of  $\rho = 2.3$ .) The closed circles are use the same Yukawa-Tsuno method and with this method the slope is  $\rho^+ = -0.82 \pm 0.20$  compared to the authors value of  $\rho^+ = -0.30$ .

Code for this plot can be explored via [Google Colab](#)

### Replotting Figure 2 the with a Model

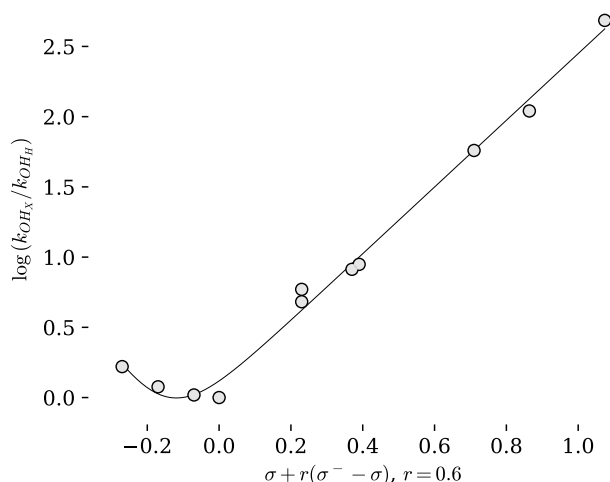
The two line fits in the plot are unsatisfying to me. It requires the researcher to bin the points into one of the two mechanisms. Should the  $X = H$  and  $X = m\text{-Me}$  rates we included with electron-withdrawing groups? That would leave only two points to define the mechanism prevalent with electron-donating groups. A better way would be to fit the plot to a model that includes both mechanisms. The model is just the addition of two Hammett plots as described in eq. 11

$$\begin{aligned}\log k_1 &= \rho_1 \sigma' + c_1 \\ \log k_2 &= \rho_2 \sigma' + c_2 \\ \sigma' &= \sigma + r(\sigma^- - \sigma) \\ \log k_{obs} &= \log(k_1 + k_2)\end{aligned}\tag{11}$$

I will use a curve fit to find best-fit parameters for  $\rho_1$ ,  $c_1$ ,  $\rho_2$  and  $c_2$  and also optimize the value of  $r$  for the best correlation constant.<sup>23</sup> The result is plotted in figure 25 on the next page. The Yukawa-Tsuno  $r$  value was optimized to be  $r = 0.60$ , very close the published value. The slope that defines the reaction that dominates the rate in the electron-withdrawing region is  $\rho_1 = 2.36 \pm 0.10$  and is very much in agreement with the authors'

<sup>23</sup> Try that with Microsoft Excel. Performing the curve fit with *Python* tools may sound challenging, but the code is already written and available on the Github repository or via Google Colab. Steal it and go.

value. However, the slope that defines the curve upwards in the electron-donating region was unable to be determined with acceptable precision. The calculated value was  $\rho_2 = -4.2 \pm 1.9$ . There is just not enough data to determine it with greater confidence (I observe only two points that describe the departure from the major linear region.)



I agree with the authors that the Hammett plot indicates a change in mechanism as substituents become electron donating. The Hammett reaction constant for the electron-withdrawing substituents,  $\rho_1 = 2.36$ , indicates a significant negative charge building up on the aniline-like nitrogen as we approach the transition state. I conclude that partial proton transfer to the hydroxide is required to make the nitrogen nucleophilic enough to attack the oxadiazole ring. The shift in slope with more electron-donating substituents shows that the more electron-rich aniline-like nitrogen can begin nucleophilic attack without proton transfer and positive charge builds up on the nitrogen as we approach the transition state. The extent of this charge is not clear but it is likely that the reaction constant for this route is negative and with a magnitude greater than 1.0. So we can say that  $\rho_2 < -1.0$  but cannot be more precise without more data.

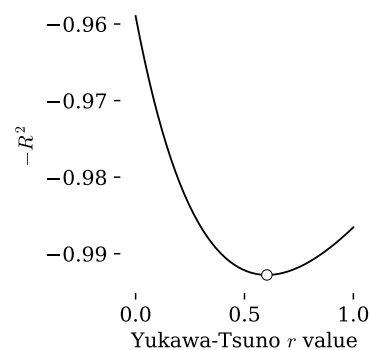
### Conclusions for the Hammett Plot for $k_{OH}$

Our initial replot of figure 2 was able to reproduce the incorrect plot presented by the authors in their paper<sup>1</sup> and revealed that there was an error in the published data table. However the use of  $\sigma^+$  parameters for a hydrazone group cannot be justified. I replotted the data using a normal Yukawa-Tsuno plot mixing  $\sigma$  and  $\sigma^-$  values and observed that the optimal mix was when we used 60 % of the resonance character of electron accepting groups ( $r^- = 0.6$ ) as shown in figure 25.

Figure 25: A replot of figure 2 in the paper with the Yukawa-Tsuno  $r$  value optimized.  $r = 0.60$  and the curve fit against the model shows that  $\rho_1 = 2.36 \pm 0.10$  (very close to the authors value of  $\rho = 2.3$ ) The second component in the rate system shows  $\rho_2 = -4.2 \pm 1.9$ . It is not a value that can be reported with any confidence.

Code for this plot can be explored via [Google Colab](#)

Figure 26: Optimizing  $r$  for the Yukawa-Tsuno plot



The plot shows a curving line that indicates a change in mechanism. So we have two different reaction mechanisms, one with a positive charge developing in the TS and one with a negative charge. Finally, I curve fit the data against a model for two simultaneous pathways. The best fit was obtained with  $r^- = 0.6$ . The reaction constant for the electron-withdrawing region was  $\rho = 2.4$ . The reaction constant for the electron-donating region was unable to be determined with sufficient precision but it is negative in value and much larger in magnitude than the value reported in the paper.

### Replotting Figure 3 Again

Let us replot figure 3 using the  $k_{H_2O}$  values determined using curve fits of the pH-rate profiles and collected in table 5 on page 16.

The best value for  $r^-$  was found to be 0.38. The plot is virtually identical to figure 10 on page 10, which used the authors data. I believe that a simple Yukawa-Tsuno plot adequately explains the system. The fit in the electron-donating side of the plot is not as perfect but the theoretical underpinnings of this approach are much more defensible.

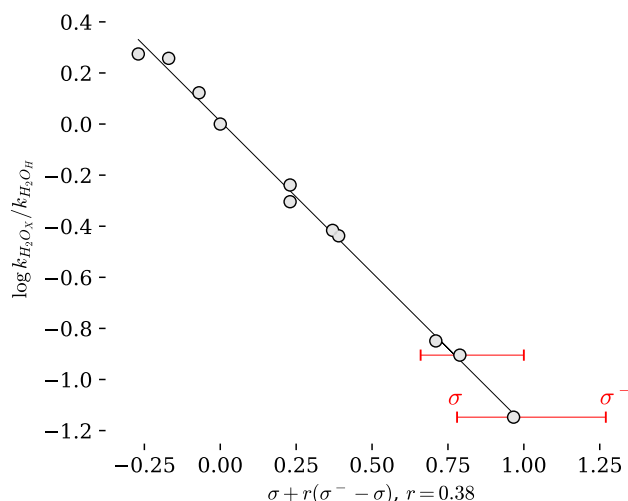


Figure 27: A replot of figure 3 in the paper as a standard Yukawa-Tsuno plot with  $r^- = 0.38$ . The slope is  $\rho = -1.18 \pm 0.02$ .

Code for this plot can be explored via [Google Colab](#)

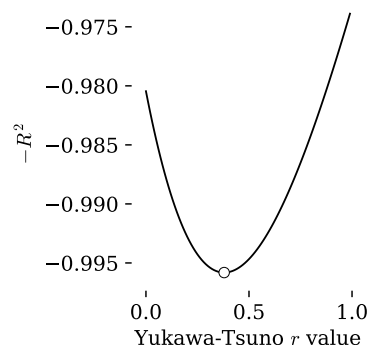


Figure 28: Optimization curve for  $r^-$  in the Yukawa-Tsuno plot presented in figure 27.

### Replotting Figure 4 Again

The same replot of figure 4 in the data using the  $k_H/K_a$  values from table 5 gives the same results. Our curve fits tracked well with the authors reported rate data. I used a normal Yukawa-Tsuno plot and observed a good line fit when  $r = 0.30$

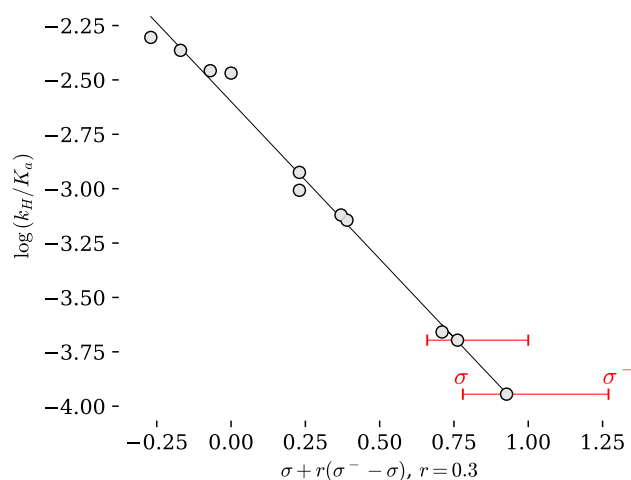


Figure 29: A replot of figure 4 in the paper as a standard Yukawa-Tsuno plot with  $r^- = 0.30$ . The slope is  $\rho = -1.45 \pm 0.05$ .

Code for this plot can be explored via [Google Colab](#)

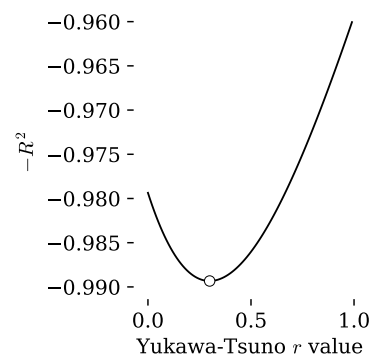


Figure 30: Optimization curve for  $r^-$  in the Yukawa-Tsuno plot.

### Plotting $K_a$ Values

For all but the p-NO<sub>2</sub> case, **11**, the curve fits reported a  $K_a$  value, however tenuous. A Hammett plot of  $pK_a$  vs  $\sigma$  shows a slope of  $\rho = 0.51 \pm 0.07$  was observed. This shows a significant effect of substituents upon the acidity of the distant oxadiazole group. The connection is an extended  $\pi$  system so a stronger connection is expected, despite the distance. You can see how the estimated standard deviation for the  $K_a$  becomes very large where the acid dissociation constants are larger and the curve in the pH-rate profile was very slight. See figure 31.

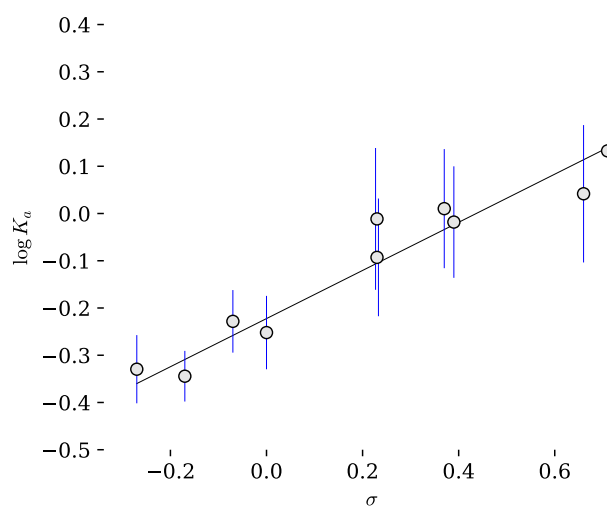


Figure 31: A Hammett plot for the  $K_a$  values determined in curve fitting the pH-rate profiles. The slope is  $\rho = 0.51 \pm 0.05$ .

Code for this plot can be explored via [Google Colab](#)

### Part 5: The Evidence and the Mechanism

That was a lot of writing and the message was that the Ingold-Yukawa-Tsuno equation was not necessary. The author provide no chemical justification for including  $\sigma^+$  parameters in the calculations for the  $x$ -axis  $\sigma$  values in figures 2, 3 and 4 of the paper. I also observed a typographical error in the data table.

So to address these issues I recalculated the rate constants in the pH-rate profile for each reactant using the data provided in the supplementary information. I followed the authors' method for obtaining data for the unsubstituted reactant, **1b**, from another paper. With these rate constants I made my own Hammett plots using the Yukawa-Tsuno method with  $\sigma^-$  parameters. The results are presented in figure 25 on page 20, figure 27 on page 21 and figure 29 on the previous page.

#### Summary of Hammett Plot results

The observations for the rate constants that define the pH-rate profile are as follows.

1.  $k_H/K_a$  – This is a second-order rate constant that defines the observed rate rate constant in the specific acid-catalyzed region. The authors used  $k_{A,R_{pH\ 1.0}}$  and we can say that  $k_{A,R_{pH\ 1.0}} = \frac{k_H}{K_a} [H^+]$  when comparing values in tables 1 and 5. Using the Yukawa-Tsuno plot we observe  $\rho = -1.45 \pm 0.05$  with the best fit observed when  $r = 0.30$ .

I conclude that the transition state involves a moderate increase in positive charge on the aniline-like nitrogen with a small amount of resonance participation by electron accepting *para* substituents.

2.  $k_{H_2O}$  – This is a first-order rate constant that defines the observed rate rate constant in the uncatalyzed region. The authors used  $k_{A,R_{pH\ 3.8}}$  and we can say that  $k_{A,R_{pH\ 3.8}} = k_{H_2O}$  when comparing values in tables 1 and 5. Using the Yukawa-Tsuno plot we observe  $\rho = -1.18 \pm 0.05$  with the best fit observed when  $r = 0.38$ .

I will make the same conclusion as above: that the transition state involves a moderate increase in positive charge on the aniline-like nitrogen with a small amount of resonance participation by electron accepting *para* substituents.

In both of the  $k_H/K_a$  and  $k_{H_2O}$  mechanisms the co-solvent, water, is the general base. The nucleophilic attack becomes well developed, generating positive character at the aniline-like nitrogen before the proton is substantially transferred. We could say that the transition state displays more nucleophilic attack than proton transfer.

3.  $k_{OH}$  – This is a second-order rate constant that defines the observed rate rate constant in the general base-catalyzed region. The authors used  $k_{A,R_{pH\ 11.5}}$

and we can say that  $k_{A,RpH\ 11.5} = \frac{K_W}{[H^+]} k_{OH}$  when comparing values in tables 1 and 5. Using the Yukawa-Tsuno plot we observe two slopes but can only have confidence in one set of parameters. One mechanism shows  $\rho = 2.36 \pm 0.10$  and the other shows  $\rho < -1$  ( $\rho = -4 \pm 2$ ). The best fit observed when  $r = 0.6$ .

The mechanism that predominates with electron-withdrawing substituents ( $\rho = 2.36 \pm 0.10$ ) indicates a significant negative charge forming on the aniline-like nucleophile in the transition state. The higher  $r$  value indicates that the lone pair on the aniline-like nitrogen has not changed as much as in the previous two cases. We can say that the nucleophilic attack is not as extensive as in case 1 and 2 above and that the proton transfer is extensive, resulting in the increase in negative character. We could say that the transition state has more proton transfer than nucleophilic attack.

The other mechanism that is taking over when substituents become more electron-donating seems to have a similar transition state structure as case 1 and 2. The hydrazone proton is no longer acidic enough to be transferring to the hydroxide without the nucleophilic attack preceding it. We could say that this transition state has more nucleophilic attack than proton transfer.

### Acid and Base Catalysis

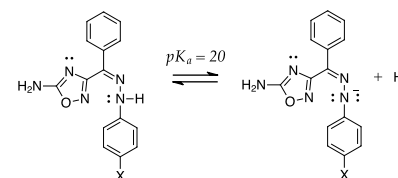
The authors present a reaction that features both acid and base catalysis. Let us first consider the base catalysis involving transfer of the hydrazone proton to a base. The authors provide evidence that this base catalysis is general. The proton is transferred as part of the rate-determining step, the nucleophilic attack of the hydrazone nitrogen upon the oxadiazole ring.

The  $pK_a$  of a phenyl hydrazone is about 20.<sup>24</sup> So hydroxide ion (the  $pK_a$  of water is 15.7) is not strong enough to remove the proton. As the nucleophilic attack proceeds, the aniline-like nitrogen will be pushing electrons density into the oxadiazole ring as the new bond forms. The in-progress nucleophilic attack will make the aniline-like nitrogen a much stronger acid and the proton can be transferred during the nucleophilic attack. Vice versa, the aniline-like nitrogen is not a strong nucleophile. The simultaneous proton transfer will increase electron density and strengthen the nucleophilic character of the nitrogen as the reaction approaches the transition state.

Another way to say this is that proton transfer will make the nitrogen a stronger nucleophile and the nucleophilic attack makes the nitrogen a stronger acid. There is no base in the system strong enough to accept the proton unless the transfer is simultaneous with the ring rearrangement.

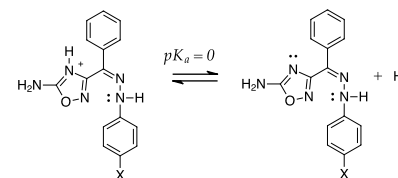
Now let us consider the acid catalysis. The  $pK_a$  of a protonated 1,2,4-oxadiazole ring is reported to be  $-0.30$ ,<sup>25,26</sup> which is very close to the values found in the curve fits of the experimental pH-rate profiles. The  $pK_a$  of hydronium is  $-1.57$ . So the hydronium acid is strong enough to completely

Figure 32: Acid equilibrium for the hydrazone group



<sup>24</sup> The ACS organic chemistry division provides several tables of collected  $pK_a$  values. <https://organicchemistrydata.org/hansch/research/pka/>

Figure 33: Acid equilibrium for the oxadiazole group



<sup>25</sup> "1,2,4-Oxadiazoles." L.B. Clapp, *Advances in Heterocyclic Chemistry*, 1976, 20, 65-116, [https://doi.org/10.1016/S0065-2725\(08\)60852-1](https://doi.org/10.1016/S0065-2725(08)60852-1)

<sup>26</sup> Again, the ACS collection of data is a wonderful source of  $pK_a$  values. <https://organicchemistrydata.org/hansch/research/pka/>



transfer a proton to the oxadiazole group. We see the proof of this when we observe the acid-catalyzed rate reaching a maximum as the reactant population becomes substantially protonated at higher concentrations of hydronium (low pH.) The protonated oxadiazole ring is much more reactive than the unprotonated ring and this species will show a higher rate of rearrangement.

### Visualizing with a More O'Ferrall-Jencks Diagram

Consider a More O'Ferrall-Jencks diagram of the situation. On one axis we will measure proton transfer from the hydrazone group and on the other we will measure progress of the rearrangement (the nucleophilic attack). These two axes follow the changes that occur in the rate determining step.

In figure 34 we see a transition state for the uncatalyzed reaction. The nucleophile attack of the rearrangement is more extensive than the proton transfer. In figure 35 on the next page is presented the specific acid catalyzed mechanism. The ts is similar, but the protonation results in an increase in the extent of the rearrangement and a decrease in protonation. The two effects are not equal as we see a ts with a greater positive charge and more extensive rearrangement.

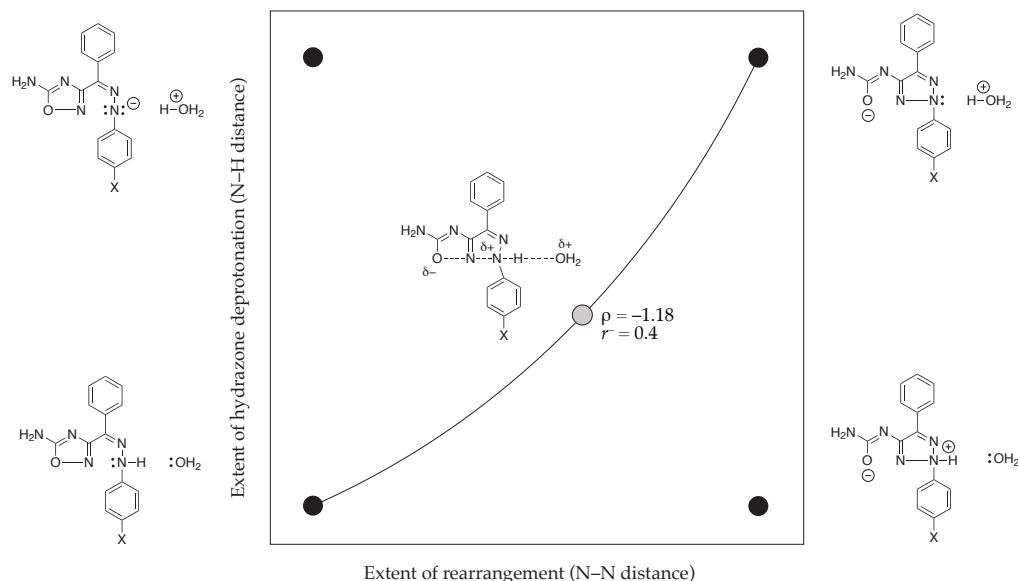


Figure 34: A More-O'Ferrall-Jencks plot for the uncatalyzed mechanism defined by  $k_{\text{H}_2\text{O}}$ .

In figure 36 on the next page we see a transition state for the general base-catalyzed mechanism. When the hydrazone proton is a strong enough acid to begin transferring to the hydroxide we see extensive proton transfer and less rearrangement in the transition state structure. The nitrogen nucleophile develops a significant negative charge. With electron-donating substituents we see the beginnings of a curve in the Hammett plot indicating a switch to a different transition state structure. This is likely similar to the ts structure in the case of the uncatalyzed mechanism where significant rearrangement was

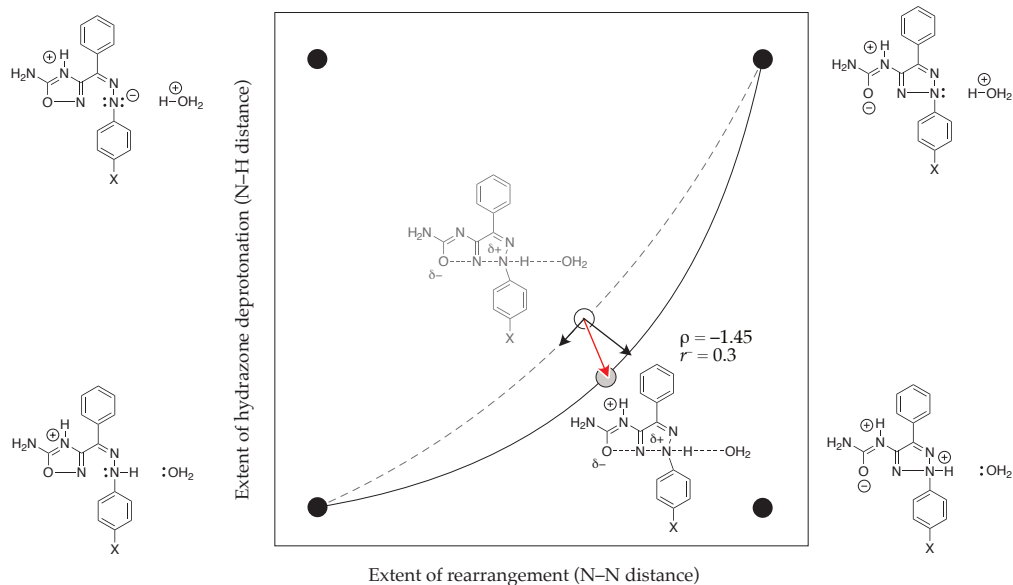


Figure 35: A More-O'Ferrall-Jencks plot for the specific acid catalysis mechanism defined by  $k_H/K_a$ .

required before the proton was acidic enough to transfer. Electron-donating substituents will decrease the acidity of the nitrogen nucleophile. There is not enough information to establish the  $\rho$  value with confidence, but it must be significantly negative to result in the curved Hammett plot that was observed.

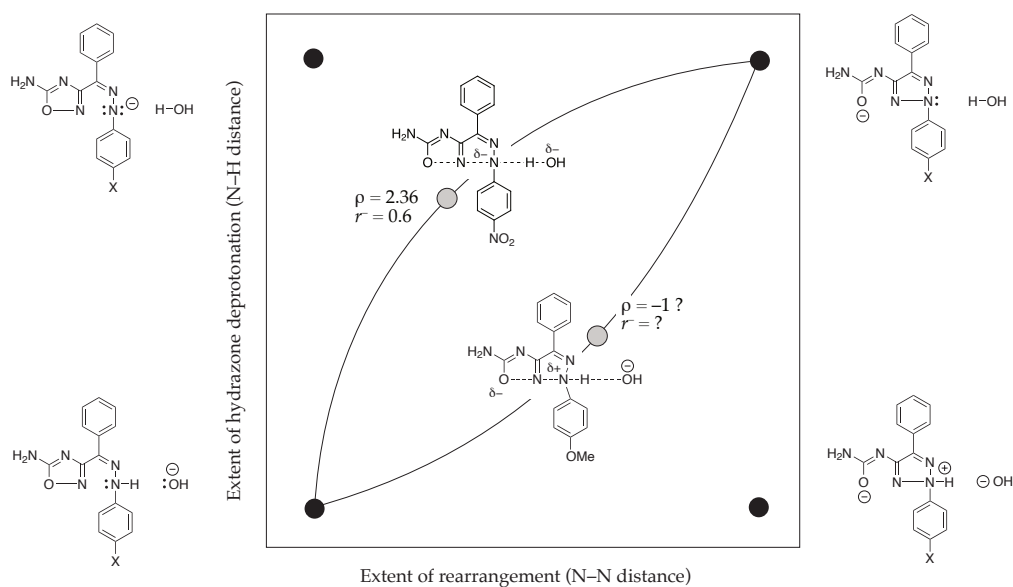


Figure 36: A More-O'Ferrall-Jencks plot for the general base-catalyzed mechanism defined by  $k_{OH}$ .

### *Final (Yes Finally!) Conclusion*

Wow, this went on way too long. The document above represents more of a stream of consciousness as I worked through all the data inconsistencies and figured out how to perform the 2-D optimization of  $r^+$  and  $r^-$  in the Ingold-Yukawa-Tsuno plots. I learned a lot of *Python* while doing this (this was my first attempt at using *Python*, although I have returned to this document and improved the code and plots several times since.)

Here are my conclusions based on my own analysis of the data. I agree with the conclusions of the authors despite my picky comments about the data and plot choices.

1. The reaction requires general base-catalysis. The proton on the aniline-like nitrogen of the hydrazone is transferred as the ring rearrangement proceeds.
2. The structure of the transition state changes depending on the relative strength of the acidic proton on the hydrazone and the base.
  - When the base is not strong enough, the general catalysis occurs later along the reaction coordinate than the rearrangement. The positive charge accumulating on the nitrogen increases the acidity enough for the required proton transfer to proceed.
  - When the base is strong enough, the general catalysis occurs earlier along the reaction coordinate than the rearrangement. The negative charge accumulating at the nitrogen increases its nucleophilicity so that the rearrangement can proceed.
3. Specific acid catalysis can occur in strong acid conditions where the oxadiazole ring is protonated. The general base requirement remains with water acting to accept the proton in the rds.
4. The use of  $\sigma^-$  parameters for electron-accepting substituents is appropriate. The lone pair of the aniline-like nitrogen can donate electrons via resonance to electron accepting *para* substituents such as p-NO<sub>2</sub> and p-CN (reactants 1j and 1l) when it can occupy a p-orbital in an sp<sup>2</sup> centre. We begin with an sp<sup>2</sup> nitrogen in a hydrazone and end with an sp<sup>2</sup> in a triazole. In the transition state the nitrogen is four-coordinate and likely tetrahedral. So it is reasonable to expect only partial contribution of the resonance effect.
  - The Yukawa-Tsuno plots show  $r = 0.4$  for the uncatalyzed system ( $k_{H_2O}$ ) where the lone pair is substantially involved in the rearrangement and no longer in the perfect trigonal-planar configuration requires to be sp<sup>2</sup>.

- In the case of the specific acid catalysis ( $k_H/K_a$ ), we observe that  $r = 0.3$ . The rearrangement may be more developed in the ts here, resulting in less  $sp^2$  character for the nitrogen. The proton transfer must be less extensive and to a greater degree as there appears to be slightly more positive charge on the nucleophile.
- In the case of the specific base catalysis ( $k_{OH}$ ), we see a significant negative charge on the nitrogen and the  $r^- = 0.6$  value indicates that the rearrangement has not progressed as far and the nucleophilic centre has more  $sp^2$  characteristics.

### Criticism

Sometimes I am negative and pedantic because I can't help myself, but this is not one of those times. The pH-rate profile data backs up the conclusions of the authors, but the use of  $\sigma^+$  parameters for an aniline nitrogen centre is not appropriate. The small  $r^+$  value presented by the authors of  $r^+ = 0.10$  and 0.08 should tell the reader that  $\sigma^+$  values are not at play here. The use of  $\sigma^+$  in the base-catalyzed Hammett plot was also without any foundation. Yes, it makes for a more dramatic departure from the linear, but the authors do not give any reason for using  $\sigma^+$  values here.

In that 1978 paper there is a reference to the method for making the IYT plot.<sup>27</sup> It provides Wepster's argument for an equation that measures both resonance donation and acceptance together. However, I would expect that only one of the  $r$  values will ever be a significant in a given mechanism. Spinelli's own values bear this out with the  $r^+$  value being tiny

The fact that the authors use Wepster's  $\sigma''$  values means that the resonance donation across the benzene ring that is included in Hammett's  $\sigma$  values is not included. Mixing back in about 10 % of the value of  $\sigma^+$  just returns us to values close to Hammett's. Again, I think that using the more common Hammett system would be better.

<sup>27</sup> "Substituent effects. II. Calculation of mesomeric p-interaction energies and comments on the relation between equations obtained by Hine and by Yukawa and Tsuno." B.M. Wepster, *J. Am. Chem. Soc.*, 1973, 95, 102-104. <https://doi.org/10.1021/ja00782a016>

### Final Thoughts

I chose this paper<sup>1</sup> because it had examples of pH-rate profiles and Hammett plots and data tables published in the supplementary material. It provides good practice in data analysis and mechanistic interpretation. And it led us into the weeds – deep into the weeds. I hope you enjoyed the trip.

It might be fun to use the IYT approach but with the more commonly used  $\sigma_I$  polar substituent parameters rather than the hard-to-find  $\sigma_n$  parameters. All the data is available in the source code. I will leave that as an exercise for the reader.

In the next section I will reanalyze the data using my own best-fit results. We will see that the Ingold-Yukawa-Tsuno plots described by the authors are not correctly presented in the paper and also argue that this multi-

parameter approach is inaccurate and unnecessary.

## Appendix

Below are the data tables and plots reference in the text above.

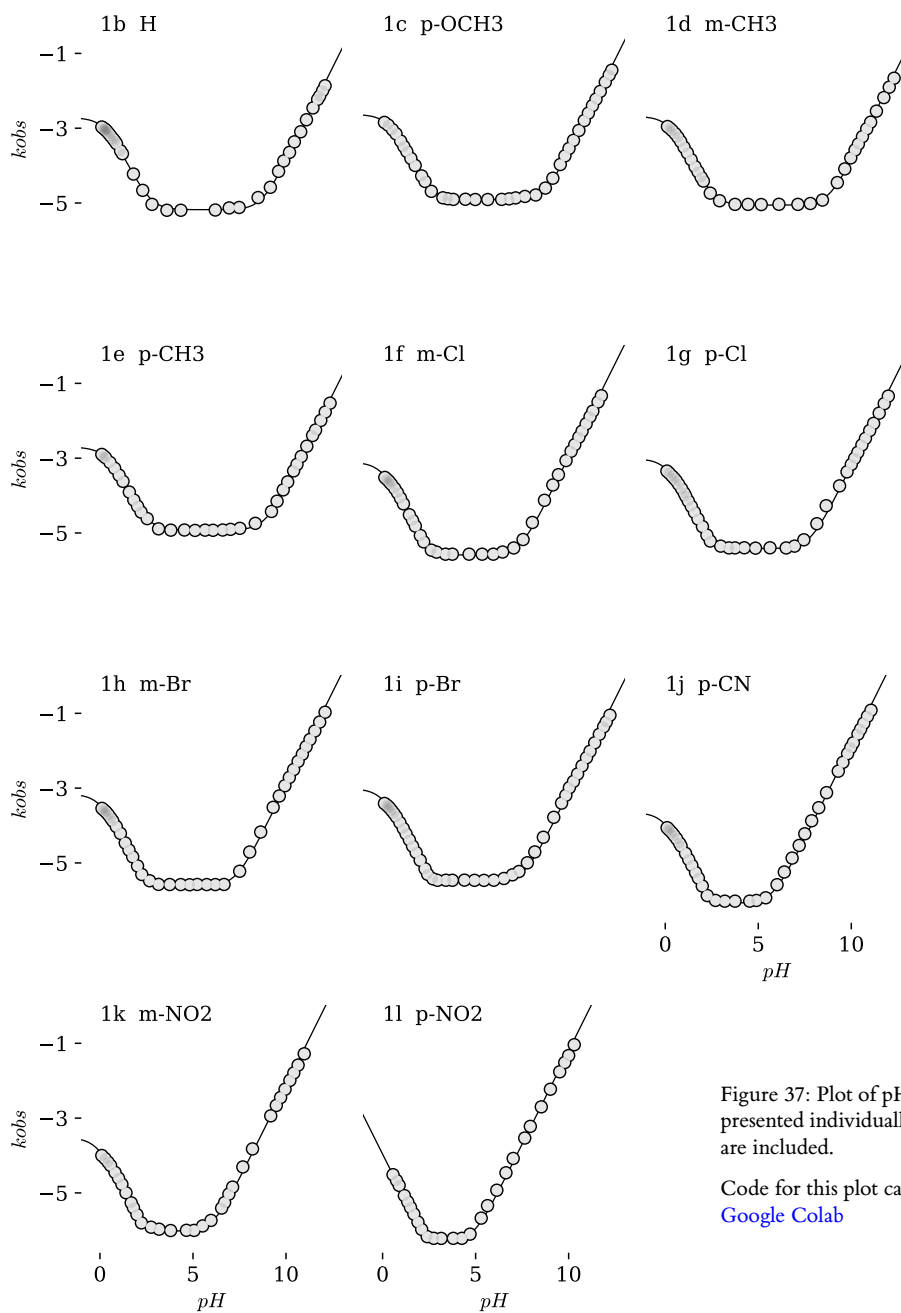


Figure 37: Plot of pH-rate profiles for 1b-l presented individually. The curve fit lines are included.

Code for this plot can be explored via [Google Colab](#)